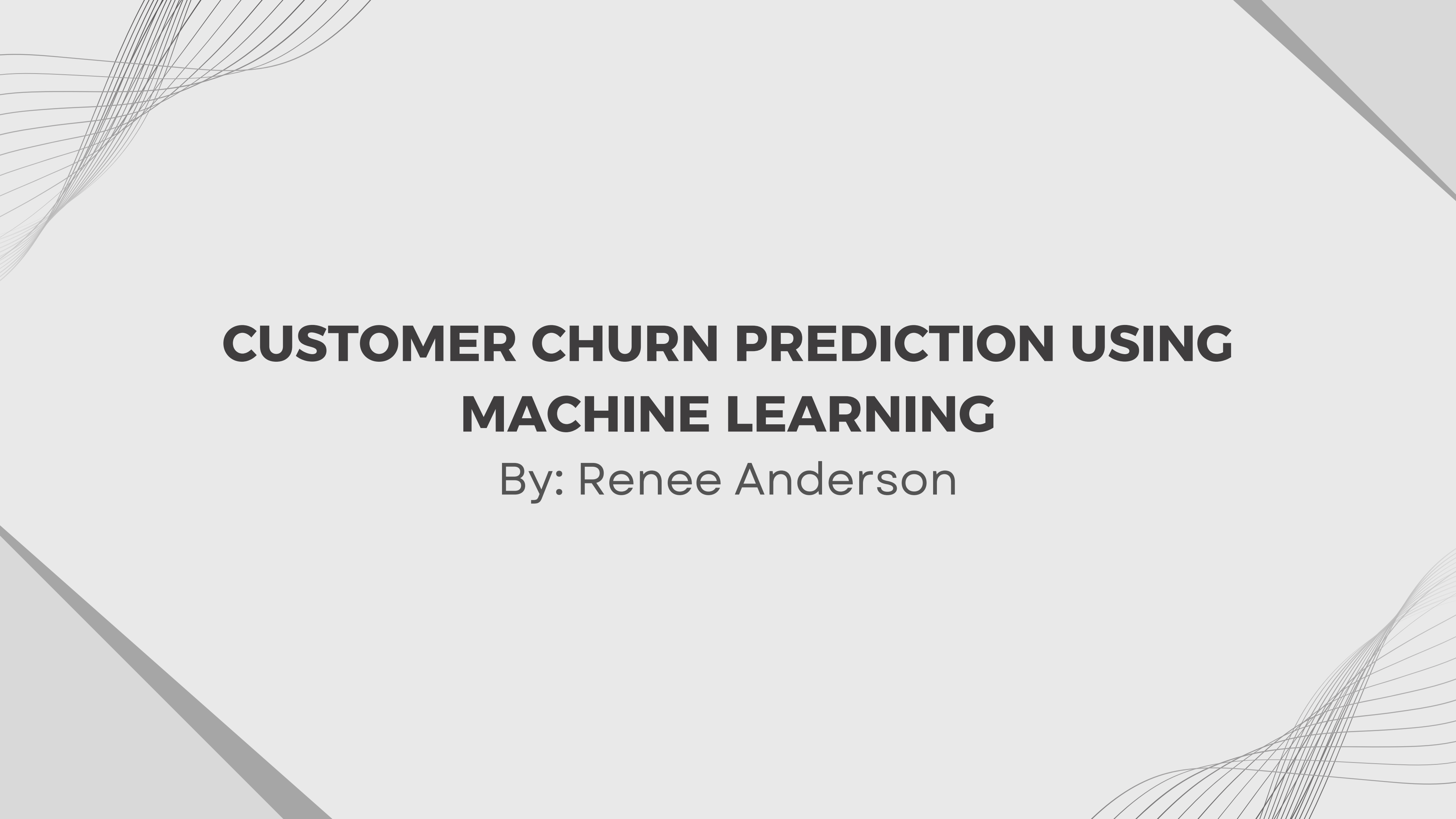
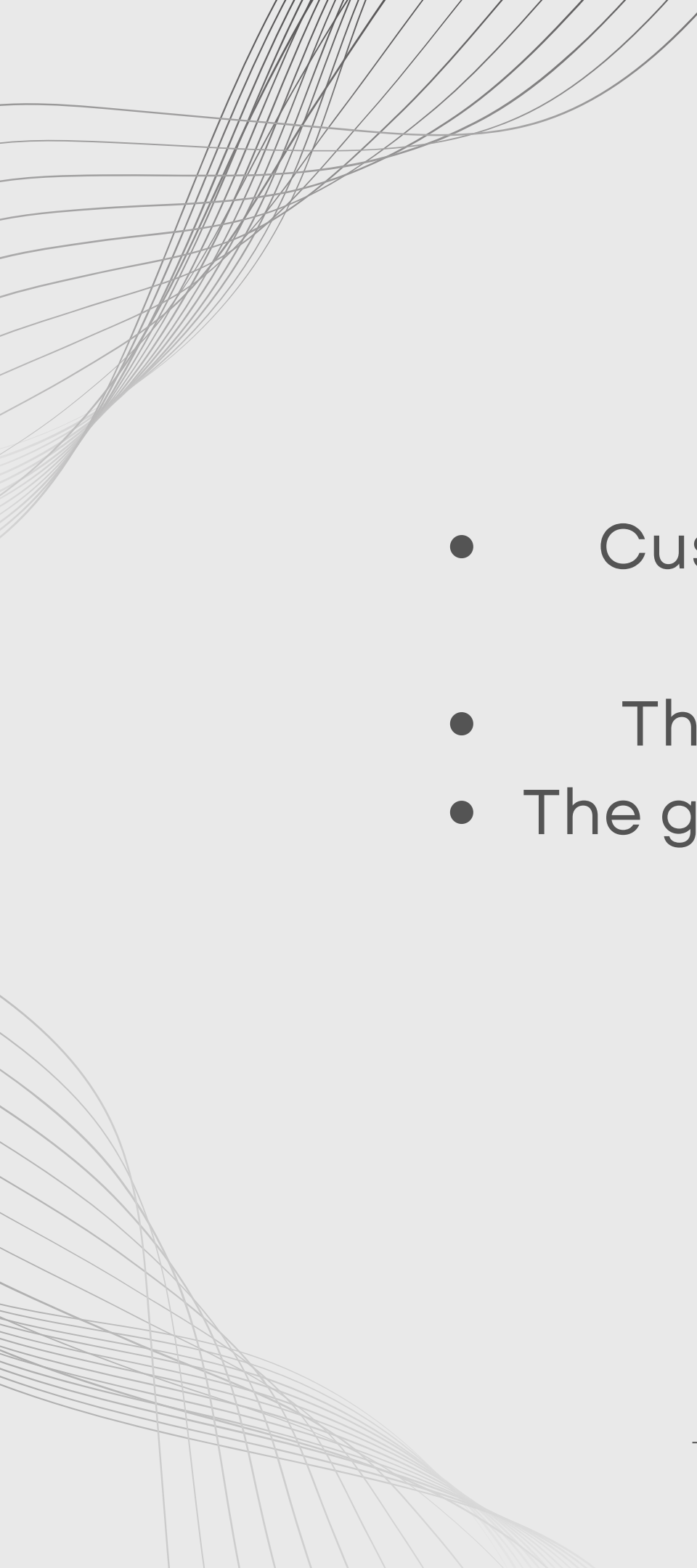




CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

By: Renee Anderson



THE PROBLEM

- Customer churn refers to when a customer leaves a company
- This results in loss of revenue and business impact
- The goal is to predict churn early using machine learning

DATASET OVERVIEW (DF.HEAD())

- Telco Customer Churn dataset (~7000 records)
- Includes customer details, charges, tenure, contracts
- Previewed dataset using df.head() to inspect structure

df.head()

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovi
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	

5 rows × 21 columns

df.head()

leLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
lo phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
No	DSL	Yes	...	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
lo phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes

DATA UNDERSTANDING

(DF.COLUMNS)

- Examined dataset columns using df.columns
- Used df.columns to list all features in the dataset
- Identified numerical and categorical features
- Helps understand how the data is organized



df.columns

```
... Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',  
         'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
         'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',  
         'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',  
         'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],  
         dtype='object')
```

DATA CLEANING

(DF.ISNULL()) 4-6

- Checked for missing values using `df.isnull().sum()`
- Converted TotalCharges to numeric format
- Removed rows with missing values
- Ensured data quality before modeling

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

DATA PREPARATION

(GET_DUMMIES) 7-8

- Converted categorical values (Yes/No) into numeric format using encoding
- Defined feature variables (X) and target variable (y)
- Prepared data for machine learning models

```
df = pd.get_dummies(df, drop_first=True)
```

TRAIN/TEST SPLIT CELL 9

- Split dataset into training and testing sets
- Training data used to build model
- Testing data used to evaluate performance

```
from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

MODELS (CELL 11-15)

Applied multiple models to compare performance:

KNN

Logistic Regression

Random Forest

XGBoost

Neural Network

```
from xgboost import XGBClassifier

xgb = XGBClassifier(use_label_encoder=False, eval_metric='logloss')
xgb.fit(X_train, y_train)

y_pred_xgb = xgb.predict(X_test)

print("XGBoost Accuracy:", accuracy_score(y_test, y_pred_xgb))

/usr/local/lib/python3.12/dist-packages/xgboost/training.py:200: UserWarning: [18:49:34] WARNING: /__w/xgboost/xgboost/src/learner.cc:782:
Parameters: { "use_label_encoder" } are not used.

   bst.update(dtrain, iteration=i, fobj=obj)
XGBoost Accuracy: 0.7739872068230277

from sklearn.neural_network import MLPClassifier

nn = MLPClassifier(max_iter=500)
nn.fit(X_train, y_train)

y_pred_nn = nn.predict(X_test)

print("Neural Network Accuracy:", accuracy_score(y_test, y_pred_nn))

Neural Network Accuracy: 0.7341862117981521
```

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

knn = KNeighborsClassifier()
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))

KNN Accuracy: 0.7690120824449183

from sklearn.linear_model import LogisticRegression

lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))

Logistic Regression Accuracy: 0.7789623312011372

from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier()
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))

Random Forest Accuracy: 0.7903340440653873
```



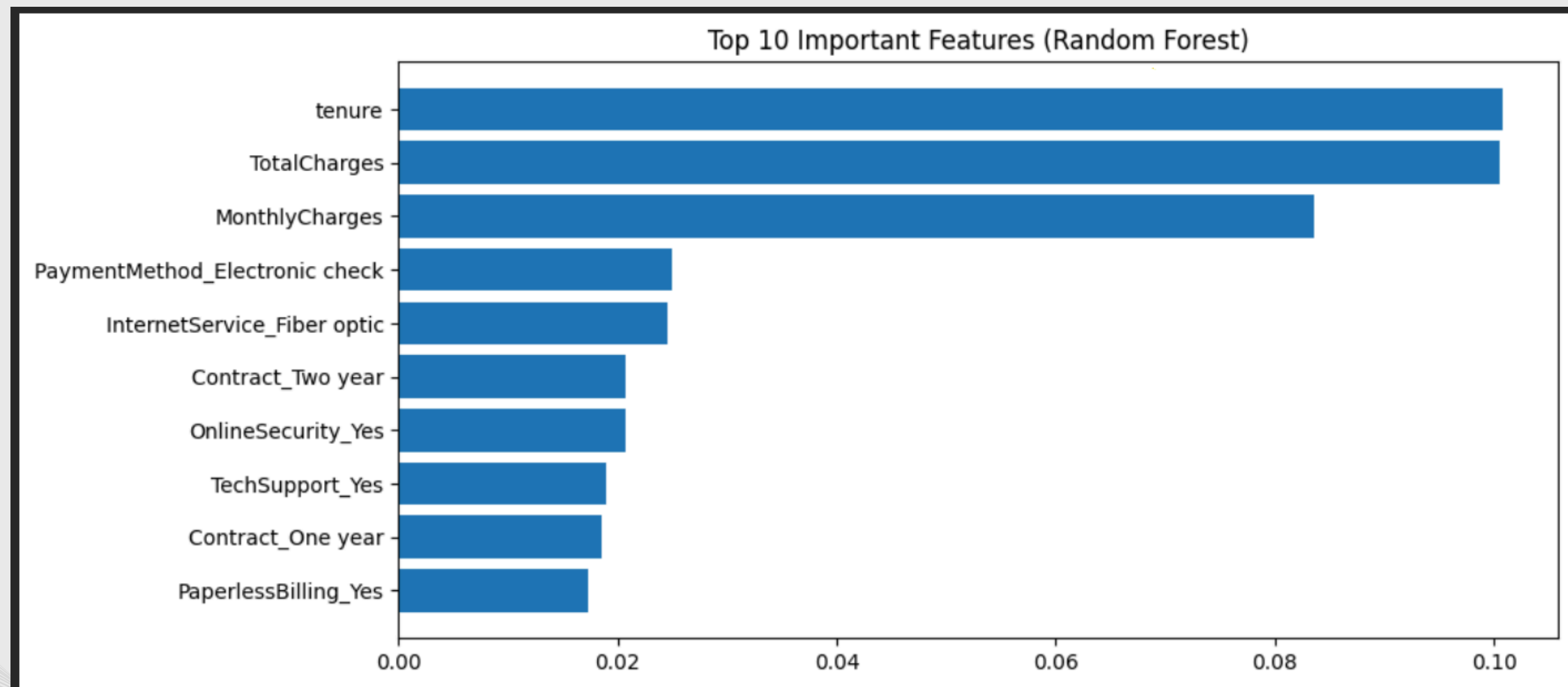
RESULTS (CELL OUTPUTS)

- KNN: 77% accuracy
- Logistic Regression: 78%
- Random Forest: 79% (best)
- XGBoost: 77%
- Neural Network: 73%

FEATURE IMPORTANCE

(CELL 16)

- Feature importance identifies key variables
- Helps explain model decision-making
- MonthlyCharges and tenure were most influential





CONCLUSION

- Random Forest performed best overall
- Model predicts customer churn effectively
- Machine learning provides valuable business insights



THANK YOU